

```
In [ ]: import pandas as pd #PARA DATOS; estructura y manipulación
import numpy as np #PARA OPERACIONES; operaciones nméricas eficientes sobre arreglo
import matplotlib.pyplot as plt #para visualización (tablas)
import requests
```

```
In [17]: df.head(3)
```

```
Out[17]:
```

|   | entity      | code | year | fertility_rate_hist |
|---|-------------|------|------|---------------------|
| 0 | Afghanistan | AFG  | 1950 | 7.248               |
| 1 | Afghanistan | AFG  | 1951 | 7.260               |
| 2 | Afghanistan | AFG  | 1952 | 7.260               |

```
In [18]: df.shape
```

```
Out[18]: (19402, 4)
```

```
In [19]: df.columns.tolist()
```

```
Out[19]: ['entity', 'code', 'year', 'fertility_rate_hist']
```

```
In [20]: df.dtypes
```

```
Out[20]: entity          str
code          str
year          int64
fertility_rate_hist  float64
dtype: object
```

```
In [21]: mexico = df[df['entity'] == 'Mexico']
japan = df[df['entity'] == 'Japan']
```

```
In [22]: mexico.shape
japan.shape
```

```
Out[22]: (77, 4)
```

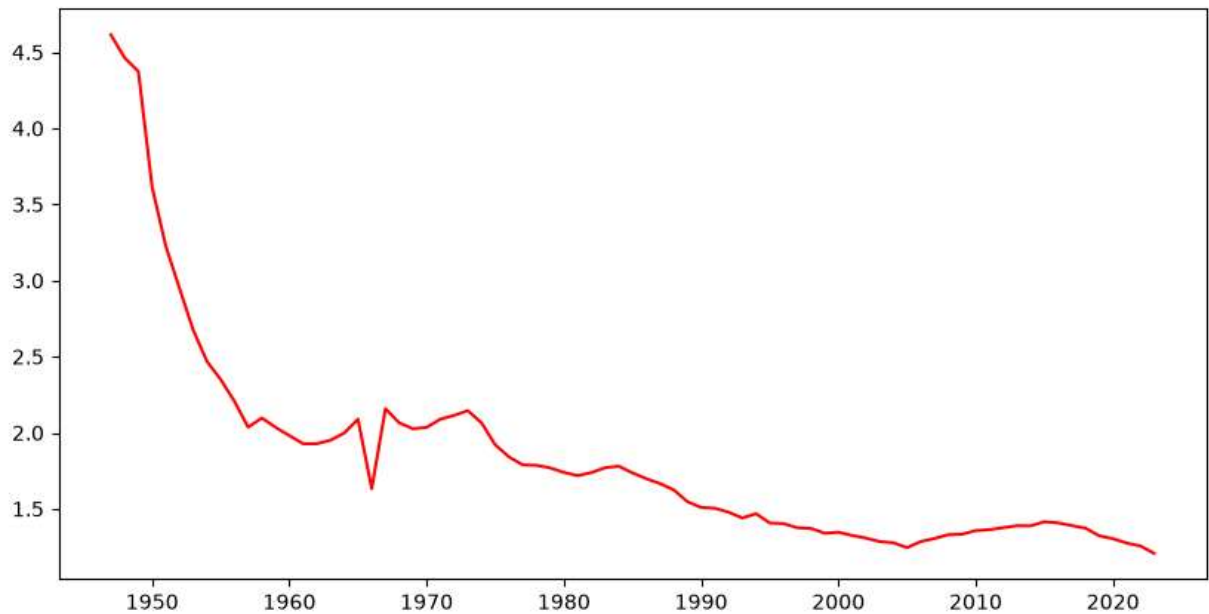
```
In [ ]: #Arreglos numpy
year_mexico = mexico['year'].to_numpy()
rate_mexico = mexico['fertility_rate_hist'].to_numpy()

year_japan = japan['year'].to_numpy()
rate_japan = japan['fertility_rate_hist'].to_numpy()
```

```
In [24]: promedio_mexico = np.mean(rate_mexico)
```

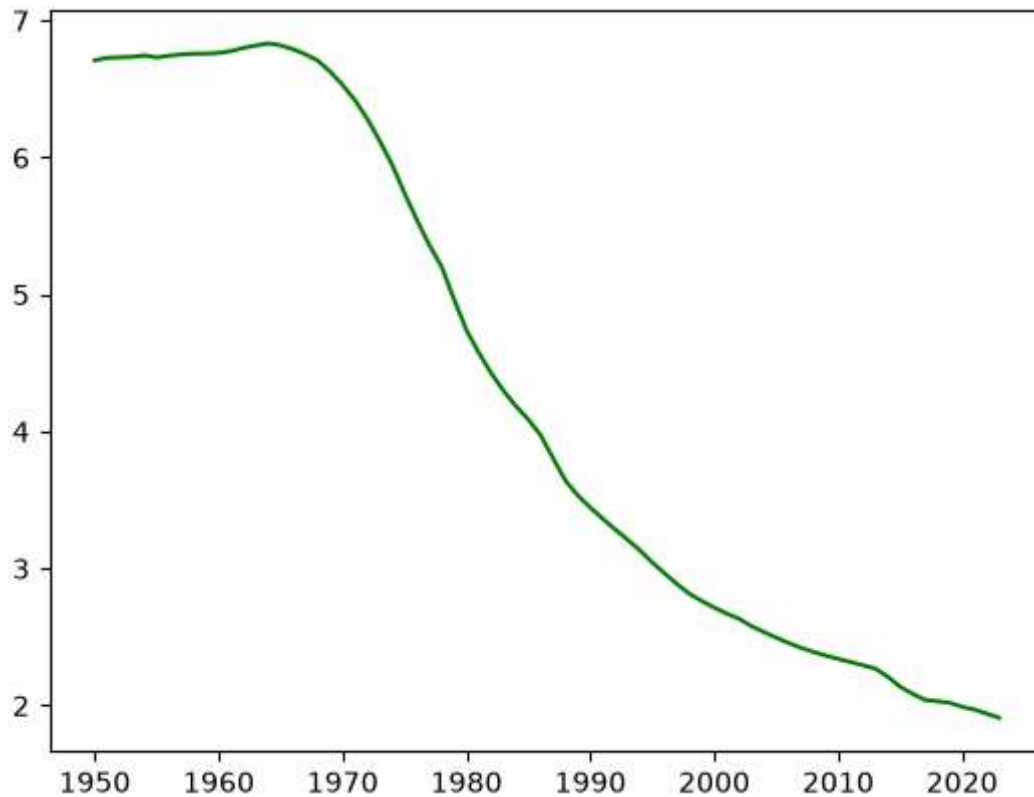
```
In [25]: plt.figure(figsize=(10, 5))  
plt.plot(  
    year_japan, rate_japan, label='Japan', color='red'  
)
```

Out[25]: [



```
In [26]: plt.plot(  
    year_mexico, rate_mexico, label='Mexico', color='green'  
)
```

Out[26]: [

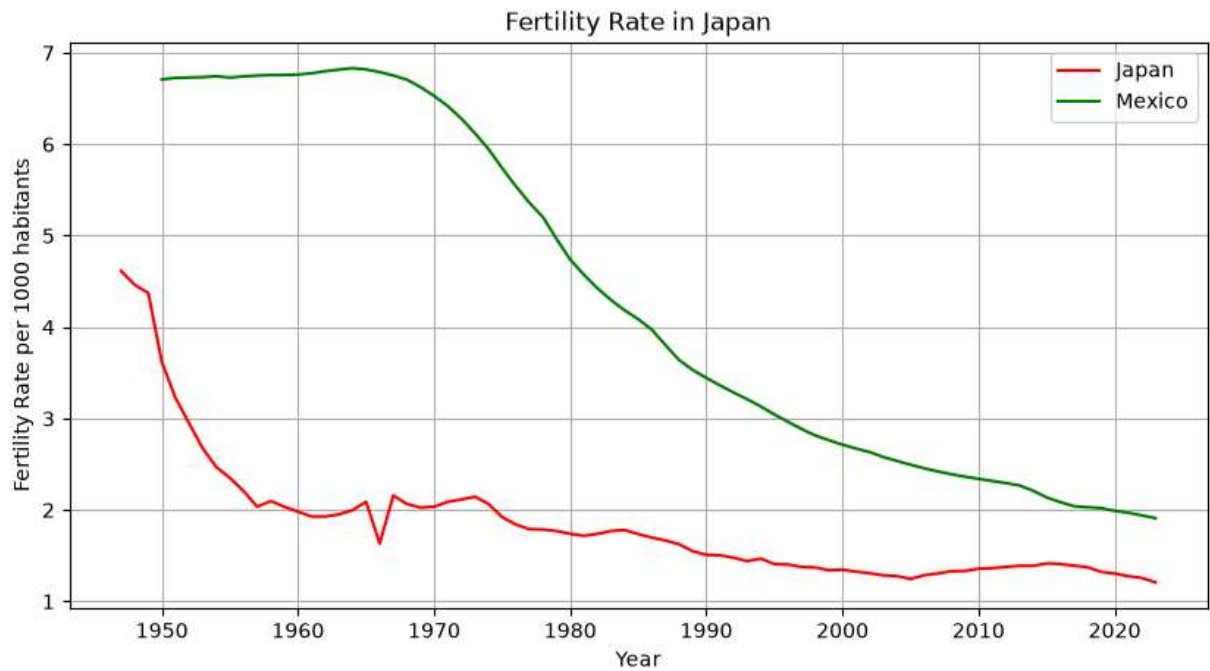


```
In [28]: plt.figure(figsize=(10, 5))
plt.plot(
    year_japan, rate_japan, label='Japan', color='red'
)

plt.plot(
    year_mexico, rate_mexico, label='Mexico', color='green'
)

plt.title('Fertility Rate in Japan')
plt.xlabel('Year')
plt.ylabel('Fertility Rate per 1000 habitants')
plt.grid()
plt.legend()
```

Out[28]: <matplotlib.legend.Legend at 0x1d2e9c13770>



```
In [30]: mexico.shape
```

```
Out[30]: (74, 4)
```

```
In [31]: japan.shape
```

```
Out[31]: (77, 4)
```

```
In [33]: mexico.head(1)
```

```
Out[33]:
```

|              | entity | code | year | fertility_rate_hist |
|--------------|--------|------|------|---------------------|
| <b>11125</b> | Mexico | MEX  | 1950 | 6.709               |

```
In [34]: japan.head(1)
```

```
Out[34]:
```

|             | entity | code | year | fertility_rate_hist |
|-------------|--------|------|------|---------------------|
| <b>8384</b> | Japan  | JPN  | 1947 | 4.614               |

```
In [47]: diferencia = rate_mexico - rate_japan
```

```
In [48]: year_japan.shape
```

```
Out[48]: (74,)
```

```
In [49]: year_mexico.shape
```

```
Out[49]: (74,)
```

```
In [60]: mascara_coincidencia_anios = year_japan >= year_mexico.min()  
mascara_coincidencia_anios
```

[illegible]

```
In [61]: year_japan2 = year_japan[mascara_coincidencia_anios]
```

```
In [63]: rate_japan2 = rate_japan[mascara_coincidencia_anios]
rate_japan2.shape
```

```
Out[63]: (74,)
```

```
In [44]: len(rate_japan)
```

Out[44]: 74

```
In [64]: diferencia_mx_jp = rate_mexico - rate_japan2
```

```
In [67]: plt.figure(figsize = (10,5))

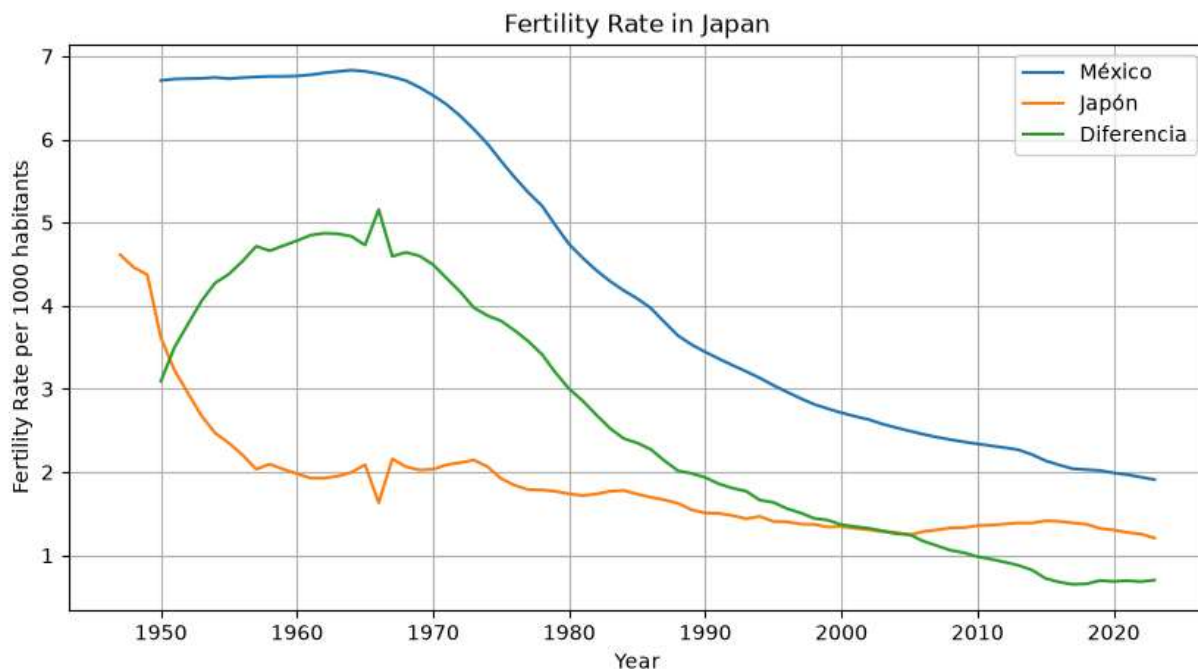
plt.plot(
    year_mexico,
    rate_mexico,
    label = "México"
)

plt.plot(
    year_japan,
    rate_japan,
    label = "Japón"
)

plt.plot(
    year_mexico,
    diferencia_mx_jp,
    label = "Diferencia"
)

plt.title('Fertility Rate in Japan')
plt.xlabel('Year')
plt.ylabel('Fertility Rate per 1000 habitants')
plt.grid()
plt.legend()
```

```
Out[67]: <matplotlib.legend.Legend at 0x1d2ed2ad550>
```



```
In [69]: #hacemos un array con los años que tienen en comun cada dataset.
common_years = np.intersect1d(
    year_mexico,
    year_japan
)
```

```
In [70]: len(common_years)
```

```
Out[70]: 74
```

```
In [71]: type(common_years)
```

```
Out[71]: numpy.ndarray
```

```
In [78]: mexico_common_years = mexico[mexico["year"].isin(common_years)]
japan_common_years = japan[japan["year"].isin(common_years)]
```

```
In [79]: mexico_rate_comYears = mexico_common_years["fertility_rate_hist"].isin(common_years)
japan_rate_comYears = japan_common_years["fertility_rate_hist"].isin(common_years)
```

```
In [80]: mex_common_years = mexico_common_years["year"].to_numpy()
jap_common_years = japan_common_years["year"].to_numpy()

mex_rate_common_years = mexico_common_years["fertility_rate_hist"].to_numpy()
jap_rate_common_years = japan_common_years["fertility_rate_hist"].to_numpy()
```

```
In [81]: print(year_japan[30])
```

```
1977
```

```
In [91]: plt.figure(figsize = (10,5))

plt.plot(
```

```
mex_common_years,  
mex_rate_common_years,  
label = "México"  
)  
  
plt.plot(  
    japan_common_years,  
    japan_rate_commYears,  
    label = "Japón"  
)  
  
plt.title('Fertility Rate in Japan')  
plt.xlabel('Year')  
plt.ylabel('Fertility Rate per 1000 habitants')  
plt.grid()  
plt.legend()
```

```

-----
TypeError                                Traceback (most recent call last)
Cell In[91], line 9
      5     mex_rate_common_years,
      6     label = "México"
      7 )
      8
----> 9 plt.plot(
     10     japan_common_years,
     11     japan_rate_commYears,
     12     label = "Japón"

File ~\Documents\Deborah\ISSC\5to\Big Data\U1_1_probabilidad_estadistica\.venv\Lib\site-packages\matplotlib\pyplot.py:4043, in plot(scalex, scaley, data, *args, **kwargs)
    4035 @_copy_docstring_and_deprecators(Axes.plot)
    4036 def plot(
    4037     *args: float | ArrayLike | str,
    (...) 4041     **kwargs,
    4042 ) -> list[Line2D]:
-> 4043     return gca().plot(
    4044         *args,
    4045         scalex=scalex,
    4046         scaley=scaley,
    4047         **({"data": data} if data is not None else {}),
    4048         **kwargs,
    4049     )

File ~\Documents\Deborah\ISSC\5to\Big Data\U1_1_probabilidad_estadistica\.venv\Lib\site-packages\matplotlib\axes\_axes.py:1792, in Axes.plot(self, scalex, scaley, data, *args, **kwargs)
    1549 """
    1550 Plot y versus x as lines and/or markers.
    1551
    (...) 1789 (`'green'`) or hex strings (`'#008000'`).
    1790 """
    1791 kwargs = cbook.normalize_kwargs(kwargs, mlines.Line2D)
-> 1792 lines = [*self._get_lines(self, *args, data=data, **kwargs)]
    1793 for line in lines:
    1794     self.add_line(line)

File ~\Documents\Deborah\ISSC\5to\Big Data\U1_1_probabilidad_estadistica\.venv\Lib\site-packages\matplotlib\axes\_base.py:331, in _process_plot_var_args.__call__(self, axes, data, return_kwargs, *args, **kwargs)
    329     this += args[0],
    330     args = args[1:]
--> 331     yield from self._plot_args(
    332         axes, this, kwargs, ambiguous_fmt_datakey=ambiguous_fmt_datakey,
    333         return_kwargs=return_kwargs
    334     )

File ~\Documents\Deborah\ISSC\5to\Big Data\U1_1_probabilidad_estadistica\.venv\Lib\site-packages\matplotlib\axes\_base.py:504, in _process_plot_var_args._plot_args(self, axes, tup, kwargs, return_kwargs, ambiguous_fmt_datakey)
    501     x, y = index_of(xy[-1])
    503 if axes.xaxis is not None:

```

```

--> 504 axes.xaxis.update_units(x)
      505 if axes.yaxis is not None:
      506     axes.yaxis.update_units(y)

File ~\Documents\Deborah\ISSC\5to\Big Data\U1_1_probabilidad_estadistica\.venv\Lib\site-packages\matplotlib\axis.py:1902, in Axis.update_units(self, data)
      1900 neednew = self._converter != converter
      1901 self._set_converter(converter)
--> 1902 default = self._converter.default_units(data, self)
      1903 if default is not None and self.units is None:
      1904     self.set_units(default)

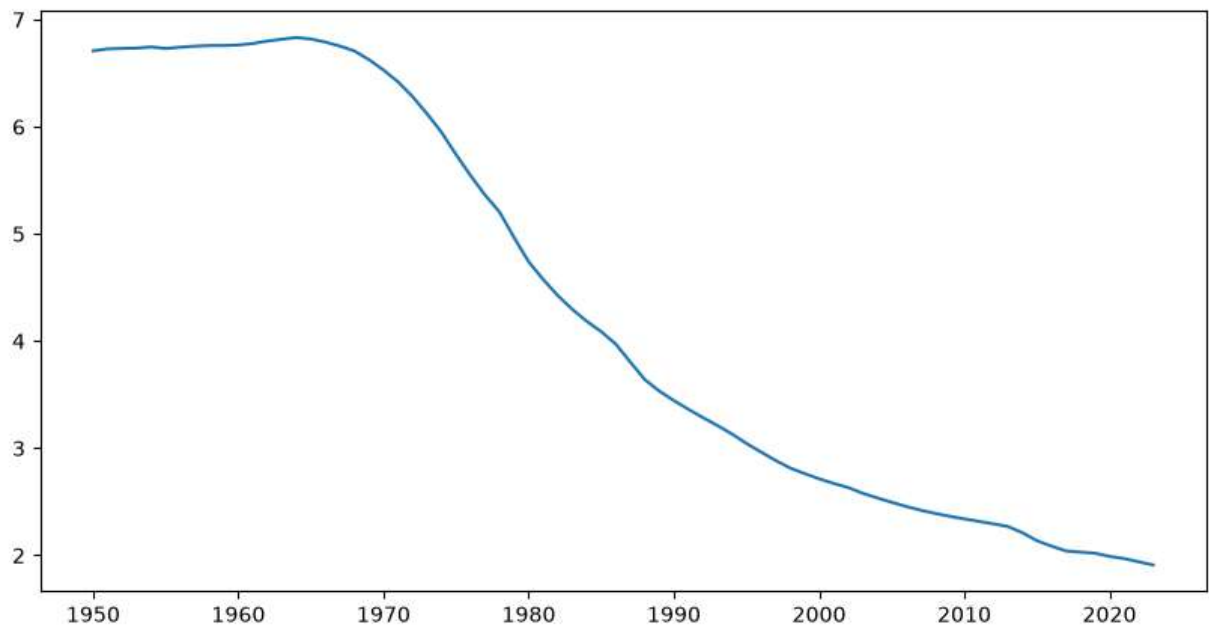
File ~\Documents\Deborah\ISSC\5to\Big Data\U1_1_probabilidad_estadistica\.venv\Lib\site-packages\matplotlib\category.py:106, in StrCategoryConverter.default_units(data, axis)
      104 # the conversion call stack is default_units -> axis_info -> convert
      105 if axis.units is None:
--> 106     axis.set_units(UnitData(data))
      107 else:
      108     axis.units.update(data)

File ~\Documents\Deborah\ISSC\5to\Big Data\U1_1_probabilidad_estadistica\.venv\Lib\site-packages\matplotlib\category.py:182, in UnitData.__init__(self, data)
      180 self._counter = itertools.count()
      181 if data is not None:
--> 182     self.update(data)

File ~\Documents\Deborah\ISSC\5to\Big Data\U1_1_probabilidad_estadistica\.venv\Lib\site-packages\matplotlib\category.py:215, in UnitData.update(self, data)
      213 # check if convertible to number:
      214 convertible = True
--> 215 for val in OrderedDict.fromkeys(data):
      216     # OrderedDict just iterates over unique values in data.
      217     _api.check_isinstance((str, bytes), value=val)
      218     if convertible:
      219         # this will only be called so long as convertible is True.

TypeError: unhashable type: 'numpy.ndarray'

```



In [82]: *##LA FORMA MÁS SENCILLA!!!*  
*#UTILIZANDO MERGE*

```
comparison_mex_jpn = pd.merge(
    mexico[["year", "fertility_rate_hist"]],
    japan[["year", "fertility_rate_hist"]],
    on = "year",
    suffixes=("_mexico", "_japon")
)
```

In [83]: comparison\_mex\_jpn.head(3)

Out[83]:

|   | year | fertility_rate_hist_mexico | fertility_rate_hist_japon |
|---|------|----------------------------|---------------------------|
| 0 | 1950 | 6.709                      | 3.615                     |
| 1 | 1951 | 6.727                      | 3.225                     |
| 2 | 1952 | 6.731                      | 2.947                     |

In [84]: comparison\_mex\_jpn["diferencia"] = (  
 comparison\_mex\_jpn["fertility\_rate\_hist\_mexico"]  
 - comparison\_mex\_jpn["fertility\_rate\_hist\_japon"] *#esto es una operacion vector*  
 )

In [85]: comparison\_mex\_jpn.head(3)

Out[85]:

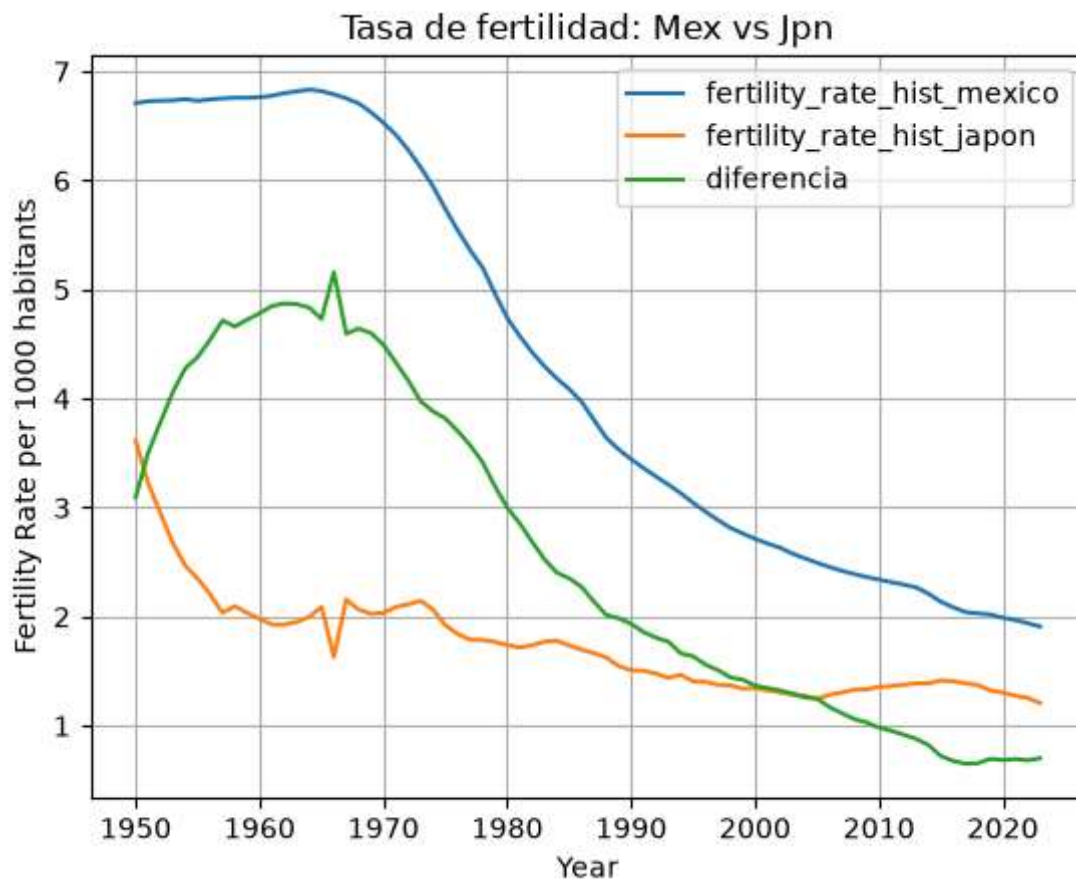
|   | year | fertility_rate_hist_mexico | fertility_rate_hist_japon | diferencia |
|---|------|----------------------------|---------------------------|------------|
| 0 | 1950 | 6.709                      | 3.615                     | 3.094      |
| 1 | 1951 | 6.727                      | 3.225                     | 3.502      |
| 2 | 1952 | 6.731                      | 2.947                     | 3.784      |

```
In [86]: plt.figure(figsize=(10,5))

comparison_mex_jpn.plot(
    x="year",
    y=["fertility_rate_hist_mexico", "fertility_rate_hist_japon", "diferencia"],
    kind="line"
)

plt.title('Tasa de fertilidad: Mex vs Jpn')
plt.xlabel('Year')
plt.ylabel('Fertility Rate per 1000 habitants')
plt.grid(True)
plt.legend()
```

Out[86]: <matplotlib.legend.Legend at 0x1d2ed4b3a10>



In [ ]: